

MOSFET – Dual N-Channel, POWERTRENCH®

60 V, 8.2 A, 17 mΩ

FDMC89521L

General Description

This device includes two 60 V N-Channel MOSFETs in a dual Power 33 (3 mm x 3 mm MLP) package. The package is enhanced for exceptional thermal performance.

Features

- Max $r_{DS(on)}$ = 17 mΩ at V_{GS} = 10 V, I_D = 8.2 A
- Max $r_{DS(on)}$ = 27 mΩ at V_{GS} = 4.5 V, I_D = 6.7 A
- Termination is Lead-free
- These Devices are RoHS Compliant

Applications

- Battery Protection
- Load Switching
- Bridge Topologies

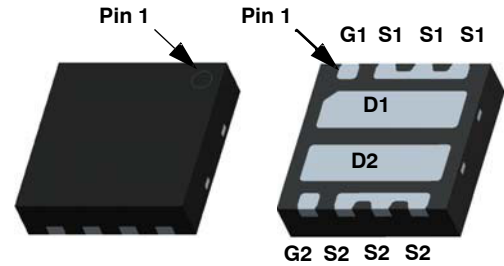
MOSFET MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Ratings	Units
V_{DS}	Drain to Source Voltage	60	V
V_{GS}	Gate to Source Voltage	±20	V
I_D	Drain Current – Continuous T_A = 25°C (Note 1a) – Pulsed	8.2 40	A
E_{AS}	Single Pulse Avalanche Energy (Note 3)	32	mJ
P_D	Power Dissipation T_C = 25°C	16	W
	Power Dissipation T_A = 25°C (Note 1a)	1.9	
T_J, T_{STG}	Operating and Storage Junction Temperature Range	–55 to +150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS

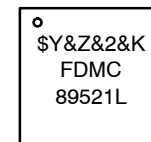
Symbol	Parameter	Ratings	Unit
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case	8.0	°C/W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1a)	65	°C/W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1b)	155	°C/W



Power 33

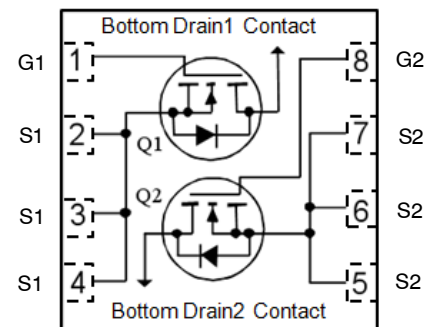
WDFN8 3x3, 0.65P
CASE 511DG

MARKING DIAGRAM



\$Y = onsemi Logo
 &Z = Assembly Plant Code
 &2 = Numeric Date Code
 &K = Lot Code
 FDMC89521L = Specific Device Code

PIN ASSIGNMENT



N-Channel MOSFET

ORDERING INFORMATION

See detailed ordering and shipping information on page 6 of this data sheet.

FDMC89521L

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Parameter	Test Conditions	Symbol	Min.	Typ.	Max.	Unit
-----------	-----------------	--------	------	------	------	------

OFF CHARACTERISTICS

BV _{DSS}	Drain to Source Breakdown Voltage	I _D = 250 μA, V _{GS} = 0 V	60			V
$\frac{\Delta BV_{DSS}}{\Delta T_J}$	Breakdown Voltage Temperature Coefficient	I _D = 250 μA, referenced to 25°C		30		mV/°C
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = 48 V, V _{GS} = 0 V			1	μA
I _{GSS}	Gate to Source Leakage Current	V _{GS} = ±20 V, V _{DS} = 0 V			±100	nA

ON CHARACTERISTICS

V _{GS(th)}	Gate to Source Threshold Voltage	V _{GS} = V _{DS} , I _D = 250 μA	1	1.9	3	V
$\frac{\Delta V_{GS(th)}}{\Delta T_J}$	Gate to Source Threshold Voltage Temperature Coefficient	I _D = 250 μA, referenced to 25°C		–6		mV/°C
r _{DS(on)}	Static Drain to Source On Resistance	V _{GS} = 10 V, I _D = 8.2 A		13	17	mΩ
		V _{GS} = 4.5 V, I _D = 6.7 A		21	27	
		V _{GS} = 10 V, I _D = 8.2 A, T _J = 125°C		20	26	
g _{FS}	Forward Transconductance	V _{DD} = 10 V, I _D = 8.2 A		28		S

DYNAMIC CHARACTERISTICS

C _{iss}	Input Capacitance	V _{DS} = 30 V, V _{GS} = 0 V f = 1 MHz		1228	1635	pF
C _{oss}	Output Capacitance			243	325	pF
C _{rss}	Reverse Transfer Capacitance			10	15	pF
R _g	Gate Resistance			0.7		Ω

SWITCHING CHARACTERISTICS

t _{d(on)}	Turn-On Delay Time	V _{DD} = 30 V, I _D = 8.2 A, V _{GS} = 10 V, R _{GEN} = 6 Ω		7.9	16	ns
t _r	Rise Time			2.1	10	ns
t _{d(off)}	Turn-Off Delay Time			18	33	ns
t _f	Fall Time			1.7	10	ns
Q _g	Total Gate Charge	V _{GS} = 0 V to 10 V, V _{DD} = 30 V, I _D = 8.2 A		17	24	nC
		V _{GS} = 0 V to 4.5 V, V _{DD} = 30 V, I _D = 8.2 A		7.9	12	nC
Q _{gs}	Gate to Source Charge	V _{DD} = 30 V, I _D = 8.2 A		3.8		nC
Q _{gd}	Gate to Drain “Miller” Charge			1.9		nC

DRAIN-SOURCE DIODE CHARACTERISTICS

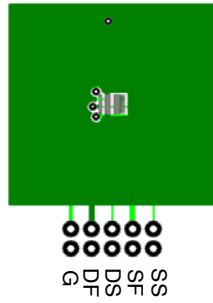
V _{SD}	Source to Drain Diode Forward Voltage	V _{GS} = 0 V, I _S = 8.2 A (Note 2)		0.85	1.3	V
		V _{GS} = 0 V, I _S = 1.6 A (Note 2)		0.75	1.2	
t _{rr}	Reverse Recovery Time	I _F = 8.2 A, di/dt = 100 A/μs		25	40	ns
Q _{rr}	Reverse Recovery Charge			11	20	nC

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

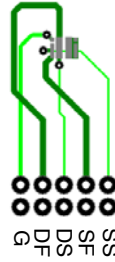
FDMC89521L

NOTES:

1. $R_{\theta JA}$ is determined with the device mounted on a 1 in² pad 2 oz copper pad on a 1.5 x 1.5 in. board of FR-4 material. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user's board design.



a. 65°C/W when mounted on a 1 in² pad of 2 oz copper



b. 155°C/W when mounted on a minimum pad of 2 oz copper

2. Pulse Test: Pulse Width < 300 μ s, Duty cycle < 2.0%.
3. E_{AS} of 32 mJ is based on starting $T_J = 25^\circ\text{C}$, $L = 1$ mH, $I_{AS} = 8$ A, $V_{DD} = 54$ V, $V_{GS} = 10$ V. 100% tested at $L = 3$ mH, $I_{AS} = 5.4$ A.

TYPICAL CHARACTERISTICS ($T_J = 25^\circ\text{C}$ unless otherwise noted)

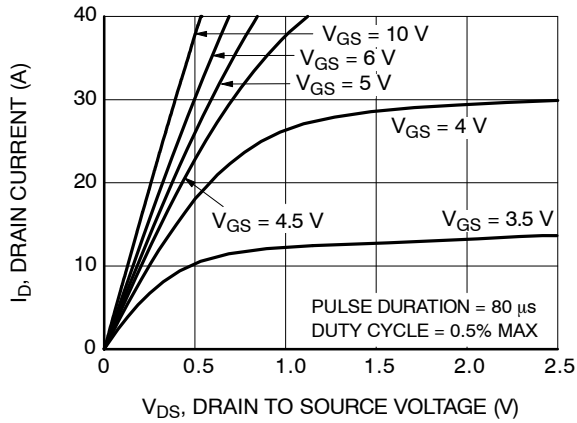


Figure 1. On-Region Characteristics

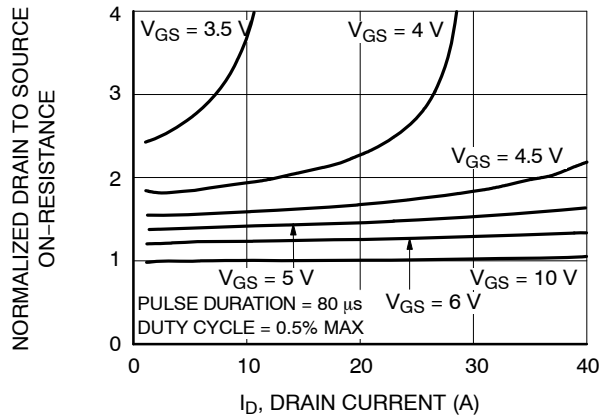


Figure 2. Normalized On-Resistance vs. Drain Current and Gate Voltage

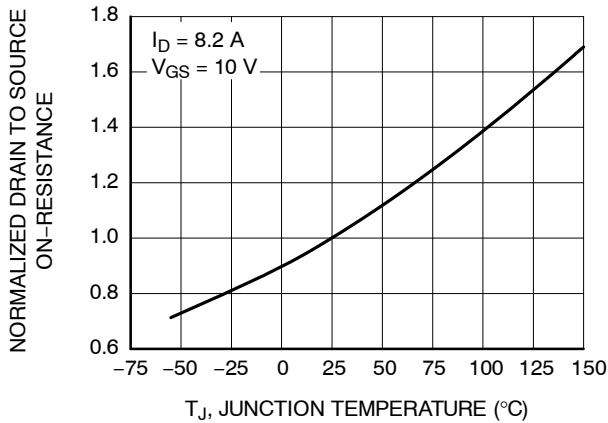


Figure 3. Normalized On-Resistance vs. Junction Temperature

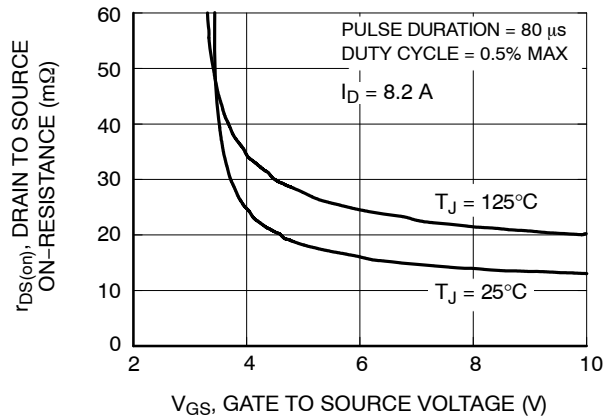


Figure 4. On-Resistance vs. Gate to Source Voltage

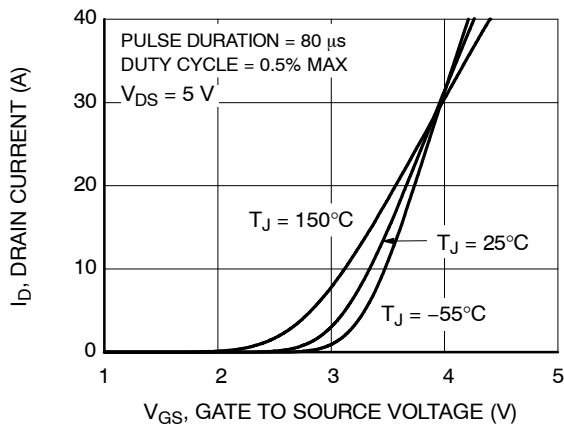


Figure 5. Transfer Characteristics

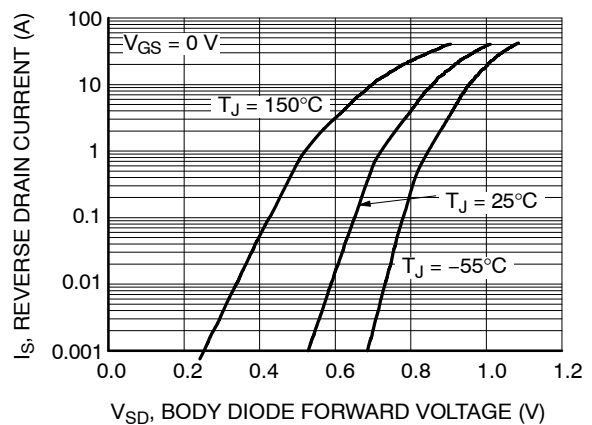


Figure 6. Source to Drain Diode Forward Voltage vs. Source Current

TYPICAL CHARACTERISTICS ($T_J = 25^\circ\text{C}$ unless otherwise noted) (continued)

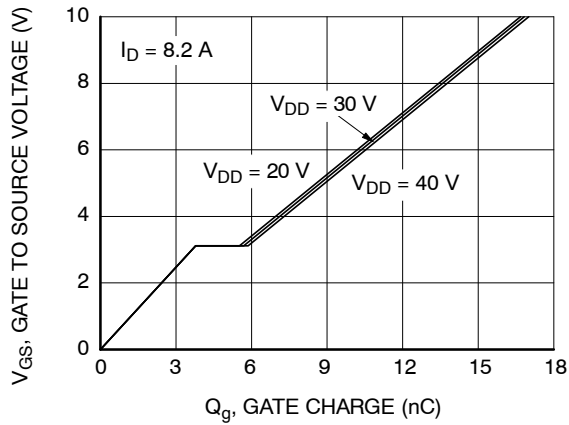


Figure 7. Gate Charge Characteristics

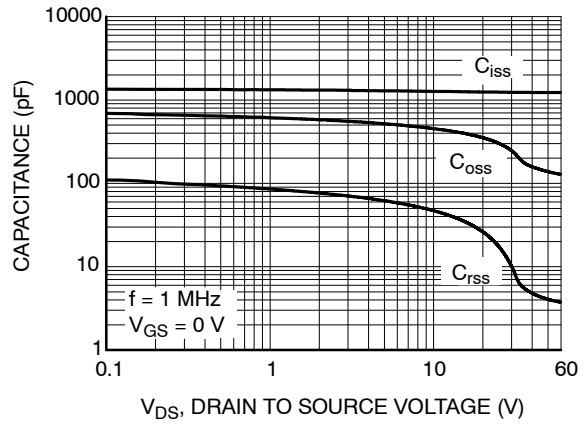


Figure 8. Capacitance vs. Drain to Source Voltage

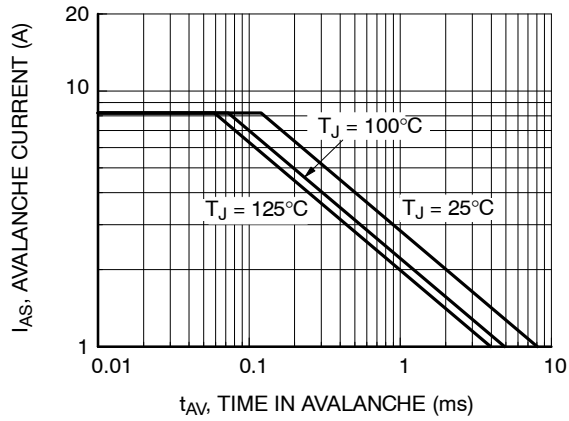


Figure 9. Unclamped Inductive Switching Capability

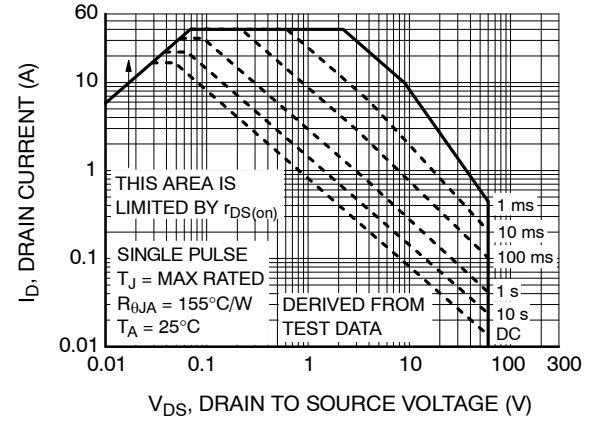


Figure 10. Maximum Continuous Drain Current vs. Case Temperature

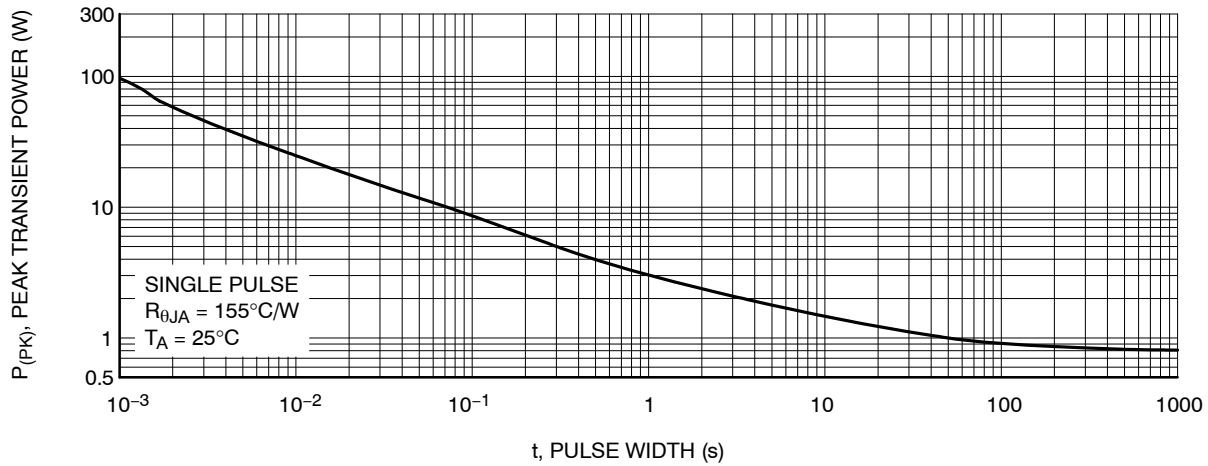


Figure 11. Single Pulse Maximum Power Dissipation

TYPICAL CHARACTERISTICS ($T_J = 25^\circ\text{C}$ unless otherwise noted) (continued)

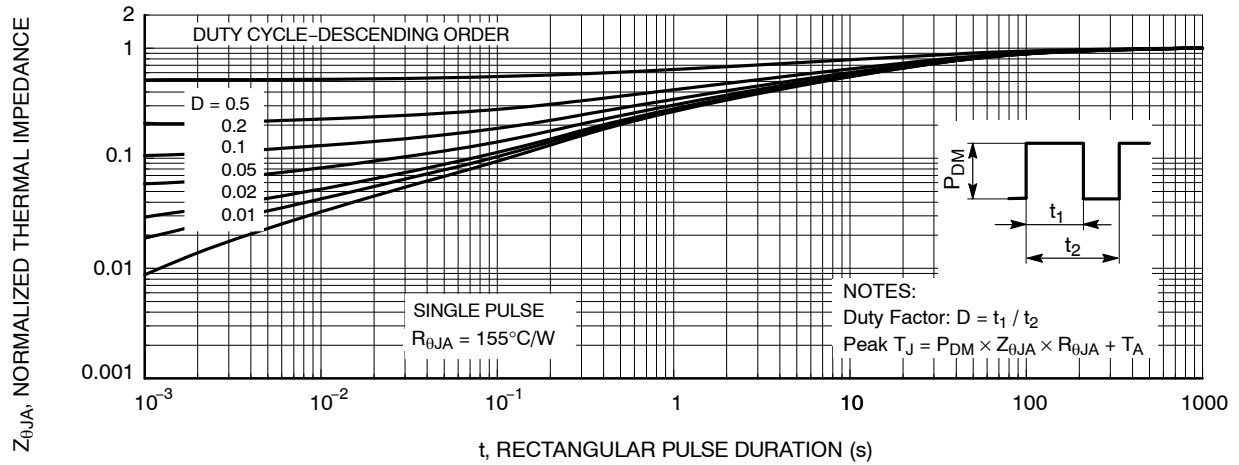
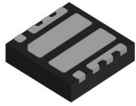


Figure 12. Junction-to-Case Transient Thermal Response Curve

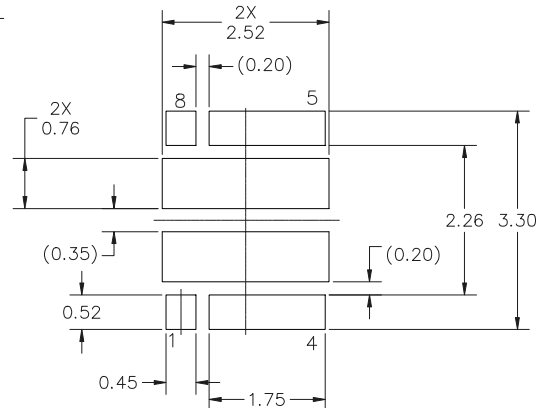
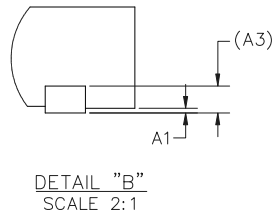
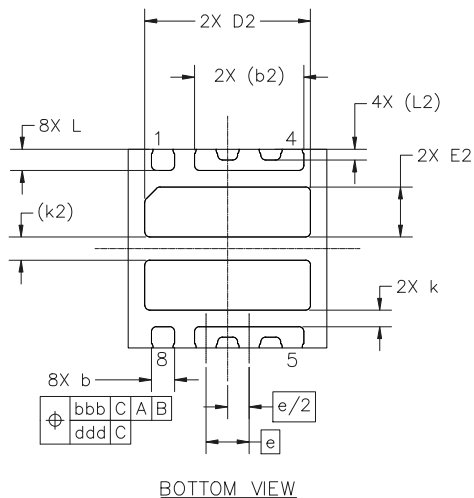
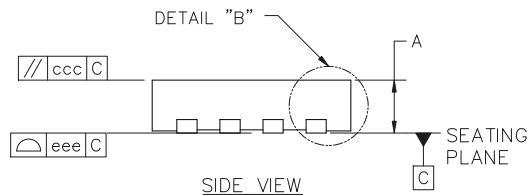
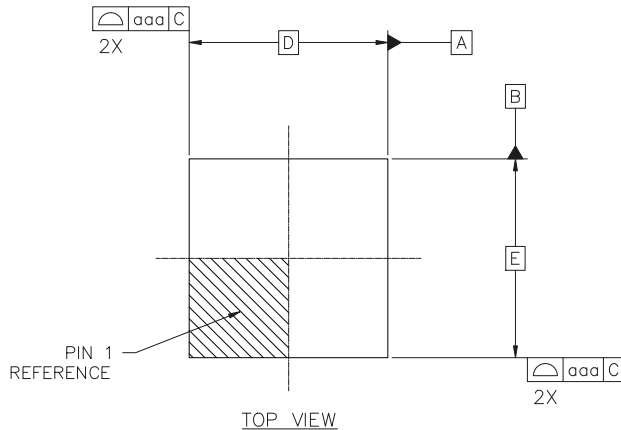
ORDERING INFORMATION

Device	Device Marking	Package Type	Shipping [†]
FDMC89521L	FDMC89521L	WDFN8 3x3, 0.65P (Pb-Free)	3000 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.


WDFN8 3.00x3.00x0.75, 0.65P
CASE 511DG
ISSUE B

DATE 15 NOV 2024



MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0.00	---	0.05
A3	0.20 REF		
b	0.30	0.35	0.40
b2	1.65 REF		
D	3.00 BSC		
D2	2.45	2.50	2.55
E	3.00 BSC		
E2	0.71	0.76	0.81
e	0.65 BSC		
k	0.22	---	---
k2	0.35 REF		
L	0.27	0.32	0.37
L2	0.16 REF		
TOLERANCE FORM AND POSITION			
aaa	0.10		
bbb	0.10		
ccc	0.10		
ddd	0.05		
eee	0.08		

NOTES:

1. DIMENSIONING AND TOLERANCING AS PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION b APPLIES TO PLATED TERMINALS AND IS MEASURED BETWEEN 0.15 AND 0.30MM FROM THE TERMINAL TIP.
4. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.

RECOMMENDED MOUNTING FOOTPRINT

- * For additional information on our Pb-Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERM/D.

DOCUMENT NUMBER:	98AON13623G	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped "CONTROLLED COPY" in red.
DESCRIPTION:	WDFN8 3.00x3.00x0.75, 0.65P	PAGE 1 OF 1

onsemi and onsemi are trademarks of Semiconductor Components Industries, LLC dba onsemi or its subsidiaries in the United States and/or other countries. onsemi reserves the right to make changes without further notice to any products herein. onsemi makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does onsemi assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. onsemi does not convey any license under its patent rights nor the rights of others.

onsemi, **Onsemi**, and other names, marks, and brands are registered and/or common law trademarks of Semiconductor Components Industries, LLC dba "**onsemi**" or its affiliates and/or subsidiaries in the United States and/or other countries. **onsemi** owns the rights to a number of patents, trademarks, copyrights, trade secrets, and other intellectual property. A listing of **onsemi**'s product/patent coverage may be accessed at www.onsemi.com/site/pdf/Patent-Marking.pdf. **onsemi** reserves the right to make changes at any time to any products or information herein, without notice. The information herein is provided "as-is" and **onsemi** makes no warranty, representation or guarantee regarding the accuracy of the information, product features, availability, functionality, or suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. Buyer is responsible for its products and applications using **onsemi** products, including compliance with all laws, regulations and safety requirements or standards, regardless of any support or applications information provided by **onsemi**. "Typical" parameters which may be provided in **onsemi** data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. **onsemi** does not convey any license under any of its intellectual property rights nor the rights of others. **onsemi** products are not designed, intended, or authorized for use as a critical component in life support systems or any FDA Class 3 medical devices or medical devices with a same or similar classification in a foreign jurisdiction or any devices intended for implantation in the human body. Should Buyer purchase or use **onsemi** products for any such unintended or unauthorized application, Buyer shall indemnify and hold **onsemi** and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that **onsemi** was negligent regarding the design or manufacture of the part. **onsemi** is an Equal Opportunity/Affirmative Action Employer. This literature is subject to all applicable copyright laws and is not for resale in any manner.

ADDITIONAL INFORMATION

TECHNICAL PUBLICATIONS:

Technical Library: www.onsemi.com/design/resources/technical-documentation
onsemi Website: www.onsemi.com

ONLINE SUPPORT: www.onsemi.com/support

For additional information, please contact your local Sales Representative at
www.onsemi.com/support/sales